
Temporal Redundancy Breaks JEPA on Satellite Image Time Series

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Abstract

1 Earth observation data is natively a time series of the same location, yet most
2 self-supervised objectives for satellite imagery treat acquisitions independently.
3 Predicting the *future* latent state of a location from its past should be the better-
4 matched objective. We show that a direct port of the JEPA objective to Sentinel-
5 2 time series **collapses**: because consecutive acquisitions of a parcel are near-
6 identical, future-latent prediction is solvable by emitting a constant (loss $\rightarrow 0$, per-
7 dimension std **[noreg]**, effective rank **[noreg]** of 512). Adding variance–covariance
8 regularization on the context embedding removes the degenerate solution. With it,
9 causal temporal prediction outperforms spatial masking by **[P1]** mIoU on PASTIS
10 crop segmentation under a frozen probe (**[P1]** vs. **[P1]**, paired t -test $p = \mathbf{[P1]}$,
11 $n = \mathbf{[P1]}$ seeds), and the margin widens from **[fewshot]** at full supervision to
12 **[fewshot]** at 1% labels. Probing frozen spatial features shows the objective makes
13 the representation phenology-aware: acquisition month decodes at **[mech]** vs.
14 **[mech]** (chance 8.3%).

15 1 Introduction

16 Contributions.

- 17 • We identify and measure a failure mode specific to satellite image time series: temporal
18 redundancy makes latent future-prediction degenerate, and we quantify the collapse via
19 effective rank and per-dimension variance.
- 20 • Under a matched-backbone, matched-budget protocol, causal temporal prediction out-
21 performs spatial masking and reconstruction/contrastive baselines on PASTIS, with the
22 advantage growing as labels shrink.
- 23 • We give direct evidence for the mechanism: the temporal objective makes *single-frame*
24 spatial features season-aware, which is the crop-discriminative signal.

25 2 Method

26 **Setup.** Given $X \in \mathbb{R}^{T \times 10 \times 128 \times 128}$ with acquisition days $d \in [1, 366]^T$, the encoder maps each
27 frame to $N=256$ tokens of width $D=512$ ($P=8$ patches). A temporal transformer aggregates frames
28 $1:t$ causally; a narrow predictor ($D_p=384$) maps the context to the EMA-target embedding of frame
29 $t+\Delta$. Day-of-year enters only as a periodic encoding; chronological order is the acquisition index.

Figure 1: Effective rank of the context embedding over pretraining on PASTIS. Without variance–covariance regularization the representation collapses to rank **[noreg]** of 512 while the prediction loss goes to zero — the degenerate solution admitted by near-identical consecutive acquisitions.

Table 1: PASTIS crop segmentation, frozen encoder, validation fold. Mean $\pm$ std over n seeds, n **stated per row**; paired t -test against temporal $\Delta=1$. All methods use an identical backbone and effective batch 192 for 100 epochs.

Objective	conv mIoU	n	p vs. temporal	parcel k -NN
Raw spectral features	[P0]	[P0]	[P0]	[P0]
Random-init encoder	[P0]	[P0]	[P0]	[P0]
MAE	[P1]	[P1]	[P1]	[P1]
SimCLR	[P1]	[P1]	[P1]	[P1]
BYOL	[P1]	[P1]	[P1]	[P1]
Temporal-order pretext	[P2]	[P2]	[P2]	[P2]
Spatial JEPA	[P1]	[P1]	[P1]	[P1]
Spatial JEPA (compute-matched, $3.5\times$ epochs)	[P1]	[P1]	[P1]	—
Temporal JEPA ($\Delta=1$)	[P1]	[P1]	—	[P1]

30 3 Temporal redundancy induces collapse

31 4 Experiments

32 5 Results

33 6 Why it wins: the representation becomes season-aware

34 7 Limitations

35 8 Conclusion

Figure 2: Test-fold mIoU at 1/5/10/100% label fractions. Temporal: **[fewshot]**; spatial: **[fewshot]**. The advantage grows from **[fewshot]** at full supervision to **[fewshot]** at 1% labels.